

ENHANCING MCLAREN F1 STRATEGY THROUGH DATA WAREHOUSING

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Executive Summary

This report is a proof of concept for the feasibility of our transactional database, the star schema derived from it, and the corresponding materialized view. Ultimately, this is fed into, PowerBI used by analysts at top F1 teams to demonstrate the relationship between factors in the management of F1 drivers and their performance in races & circuits.

Our team undertook research to understand the case study and therefore the needs of relevant stakeholders, before defining the scope of our work. This informed our decision making on our ERD and star schema.

For our ERD, understanding the importance of unique elements to the F1 environment, like real-time telemetry and the nuances of measuring driver performance were vital in ensuring that our ERD reflected real-life data flows generated from F1 races, and was able to store this data within our scope. This included solid foundations for cardinality and relationship choices, to ensure they were accurate, and that the assumptions made were reasonable and balanced feasibility and scope without omitted key considerations. On top of this, the database was normalised to 3NF, to eliminate data redundancy and improve overall efficiency of storage and queries run.

Then, we discussed the merits of all structures but landed on star schemas due to their simplicity & intuitiveness, flexibility & scalability, and the benefits from increased efficiency and performance in analyst queries and PowerBI, in a cost-conscious and highly competitive environment.

We created a fact table listing quantitative attributes measuring a driver's performance in a specific race, and 5 dimension tables providing details, the 5 W's, of each performance. We carefully considered which tables to denormalise to create these fact & dimension tables, to eliminate the need for complex join queries.

Our star schema informed the creation of four materialized views, allowing for near-instantaneous report loading and interaction in PowerBI. We demonstrate these digestible and impactful visualisations from PowerBI, from tables to charts, allowing analysts managing the performance of F1 drivers to demonstrate key insights to stakeholders without technical backgrounds.

Scope

This project focuses on measuring specific factors and outcomes for F1 drivers in one specific race.

Our database and normalized ERD models pertinent details such as driver support, race strategies, and media events, alongside performance metrics and sponsorship contracts. Subsequently, a de-normalized Star Schema will be built from the ERD to facilitate easier analysis by creating fact and dimension tables. A Materialized View will then prepare this data for a final report and visualization in Microsoft Power BI.

Stakeholder Analysis

Stakeholders are divided into two main groups: the F1 management team and technical specialists.

- For technical specialists, the project demonstrates a robust, normalised ERD that ensures data integrity and clarity for efficient database queries. This foundation is crucial for those who will implement and maintain the database system.
- For the F1 management team, many of whom are non-technical, the relational data is refined into a de-normalised Star Schema and Materialized View. This approach improves the usability of the data for analysis and business intelligence. The final presentation will use Microsoft Power BI visualisations to showcase the system's output and convince the management team of the project's viability and value.

ERD Design

The design of the ERD aims to capture track vital information that F1 management teams need in order to make the best decisions based on driver data and telemetry.

Through research of what is vital to F1 teams to make informed decisions, we identified entities that were important to add, including TELEMETRY which is extensively used by engineers during races to adjust strategies and coordinate directly with drivers to ensure cars are functioning at peak efficiency. The DRIVER_PERFORMANCE entity was also vital as it assessed the overall performance of F1 drivers during races. It includes attributes like PositionFinished and LapTimes, providing insights into the speed, efficiency and consistency of different F1 Drivers. By linking similar entities that record data to the DRIVER entity, the ERD can provide a detailed analysis over time, supporting decisions on driver development, contract negotiation and strategies.

To support and validate the importance of this entity, we conducted research on modern F1 teams aside our own such as Mercedes and RedBull, finding that teams rely heavily on constant telemetry and post-race driver data to make informed decisions on drivers. Further, we have normalised it to 3NF, eliminating redundancy and improving both the accuracy and consistency of stored data.

Assumptions

For the construction of our ERD, telemetry and performance data are assumed to be collected after specific sessions such as test sessions and races, where each driver can have multiple telemetry results over multiple sessions. This was done to justify the one-to-many relationship between both the DRIVER and TELEMETRY entities. Additionally, it's assumed that performance indicators such as EnergyManagementScore and DriverWellbeing are computed after each session and then stored independently within each separate performance record entity, linking it back to the session and driver.

Another assumption is the relationship between ROLEASSIGNMENT and PITSTOP where our model shows a one-to-many relationship from ROLEASSIGNMENT to PITSTOP and a one-to-one relationship in reverse. A PITSTOP involves multiple members where our design links each PITSTOP to only one ROLEASSIGNMENT. For this, we assumed that each PITSTOP is led by a single primary role such as a lead mechanic and that it captures the key ROLEASSIGNMENT as responsible for the PITSTOP, without adding unnecessary complexity.

Assuming that each member of the F1 management also holds only a single employment role such as MECHANIC or DRIVER supports a one-to-many relationship from the EMPLOYEE entity, where one employment type may be shared by many drivers but each DRIVER has only one employment type. These assumptions help with the proper reasoning behind the cardinalities and connectivity, also preventing many-to-many relationships and potential fan traps. Without them, the ERD would lack the necessary guidelines and would fail to reflect a data model suitable for analytics.

ERD Discussion

The finished ERD accurately represents the required entities and relationships that occur in an F1 management team and models all essential relationships and entities. Every entity was made in accordance with how a F1 management team track, evaluate and strategise based on driver performance. Entities such as the telemetry entity contains attributes such as EngineRPM, ThrottlePosition and TyreTemperature, which enables the real-time analysis of an F1's vehicle while on track. Linking telemetry data to both the DRIVER and SESSION allows the team to analyse how certain drivers perform in different circuits, track conditions and various types of sessions. Through the attribute we also included, which was StrategyNotes, allowed us to help document and note one-off conditions that may have occurred that impacted the outcome.

Looking further at the ERD model itself, each entity is well connected with correct and accurate foreign and primary keys. This is additionally supported with all relationships, connectivity and cardinality being modelled closely to the operations of an F1 management team. This model has also been normalised to 3NF to ensure the elimination of redundancy and improve the overall efficiency of storage and queries ran. Overall, the ERD model aligns itself well with the requirements of an F1 management team, forming a strong foundation for the database warehouse in SQL and the star schema.

Data Warehouse Discussion

While the transactional (OLTP) database excels at capturing real-time race data, its normalized structure is ill-suited for the complex queries required for strategic analysis. Our data warehouse bridges this gap, creating a powerful analytical asset that empowers McLaren's F1 team to dissect performance, validate strategies, and unlock competitive advantages. To support our analytical goals, we adopted a Star Schema architecture due to its direct advantages in tools like Power BI:

- Star schemas **improve Power BI performance** by minimising joins through denormalisation and consolidation of data from multiple transactional tables.

- Its **simplicity is intuitive** for business users. The "facts" (quantitative metrics) are in the center, surrounded by the business context or "dimensions". This clear separation of metrics from descriptors allows a stakeholder to easily comprehend how analysis can be isolated by driver, circuit type, vehicle setup, or race strategy.
- The model is **flexible, extensible and scalable**. New analytical factors like weather can be met by adding new or to existing dimensions without redesigning the warehouse.

Additionally, by pre-structuring data into star schemas and materialized views, we reduce computational overhead, shifting resource-intensive queries to scheduled refreshes and lowering infrastructure costs.

ERD to Star Schema Conversion

Transforming our ERD into an analytics-ready star schema involved strategic denormalisation to create consolidated, meaningful business entities.

Fact Table: *Fact_RacePerformance*, is the heart of our data warehouse, with each row informing a specific, unique event: **one driver's performance in one specific race**. It aggregates KPIs allowing analysts to perform various calculations, from simple averages to complex correlational analysis on the impact of metrics like hydration:

- **Quantitative Metrics:** *PositionFinished, Overtakes, TireDegradationRate, LapTimes*.
- **Behavioural and Conditional Metrics:** *PreRace_StressScore, PreRace_FocusLevel, RaceDayCalories, isDNF* (Did Not Finish flag).

Dimension Tables: Providing Business Context

Five considered dimension tables provide the 5 W's for each performance record:

- *DW_Driver_Dim* (**Who**): A comprehensive profile of each driver: basic information with crucial, time-sensitive wellbeing and health snapshots.
- *DW_Race_Circuit_Dim* (**Where/What conditions**): Describes the race event, including the circuit's physical characteristics and the weather conditions on race day.
- *DW_Vehicle_Dim* (**What was used**): Specifications on the car assigned to the driver for the race.
- *DW_Strategy_Dim* (**How**): Outlines the planned and actual race strategy, including pit stops, tire choice, and the manager responsible.
- *DW_F1_Date_Dim* (**When**): A standard time dimension, allowing for performance analysis across different time periods.

1. *DW_Driver_Dim*: derived from denormalising five separate entities from the ERD: *Employee*, *Driver*, *DriverWellbeing*, *HealthRecord*, *NutritionLog*, and *MentalAssessment*.
 - **Justification:** This dimension consolidates a driver's profile into a single snapshot using *ROW_NUMBER()* to extract the most recent wellbeing and health data before each race. It enables direct correlation between personal condition and performance without complex joins.
2. *DW_Race_Circuit_Dim*: formed by combining the *Race* and *Circuit* entities.
 - **Justification:** Merges *Race* and *Circuit* entities into one. This avoids a snowflake schema and reduces unnecessary joins. Attributes include *RaceName*, *WeatherCondition*, *CircuitName*, and *CharacteristicType*—giving a complete view of the racing environment.
3. *DW_Strategy_Dim*: integrates *RaceStrategy*, *Manager*, and *Employee*.
 - **Justification:** This allows for powerful analysis of strategic choices. For example, by having *TireSelection* (e.g., 'S-H-S') and *StrategyOutcome* (e.g., 'Optimal') in the same dimension, analysts can easily create a report showing which tire strategies most frequently lead to successful outcomes, providing direct, actionable feedback to the strategy team.
4. *DW_Vehicle_Dim*: direct but essential mapping from the *Vehicle* entity in the ERD.
 - **Justification:** Creating a dedicated vehicle dimension is crucial. It isolates vehicle-specific attributes (*ChassisNumber*, *EngineType*, *TopSpeed*), providing a clean and stable set of characteristics to analyse car performance and reliability over time.
5. *DW_F1_Date_Dim*: derived from the *Date_Race* attribute within the *Race* entity.
 - **Justification:** As a standard time dimension, it supports flexible temporal analysis—e.g., comparing performance across race weekends, segments of the season, or year-over-year trends.

Fact Table Construction and Data Cleansing

The *Fact_RacePerformance* table was constructed by aggregating keys and measures from a wide array of source tables, including *DriverPerformance*, *RaceAssignment*, *NutritionLog*, and *MentalAssessment*.

- **Justification:** A critical step in this process was ensuring data integrity. The *ROW_NUMBER() OVER(PARTITION BY ...)* function was used to create a unique key for each driver-race combination (*rn* = 1). This prevents duplicate records that could arise from complex joins (e.g., a driver having multiple temporary vehicle assignments). Furthermore, we used correlated subqueries to fetch specific measures like *PreRace_StressScore*, ensuring that the most relevant data point is selected for each

race event. These techniques significantly improved data quality, ensuring one-to-one mappings in the fact table and avoiding null or duplicate key violations.

Implementation of Surrogate Keys

A core principle of our data warehouse design is the use of surrogate keys. These are simple system-generated integers replacing source keys.

- **Justification:** Surrogate keys offer two major advantages. First, they are more efficient for the database engine to join than long, complex natural keys. Second, they insulate the warehouse from operational system changes. For example, if a chassis number is updated due to repairs or regulation changes, the fact table's link to the vehicle remains stable via the surrogate *VehicleKey*. This improves the long-term robustness of the data warehouse.

Materialized Views for Power BI Performance

To provide a seamless and highly responsive user experience in Power BI, we have built a layer of four Materialized Views (MVs) on top of our Star Schema. By storing the results of complex queries, MVs improve efficiency for PowerBI analysis.

Purpose and Justification

Power BI queries stable, pre-aggregated MVs, resulting in near-instantaneous report loading and interaction. This is a critical optimization strategy for handling large-scale data.

Our four MVs are designed to answer specific, high-value business questions:

1. *MV_Driver_Season_Summary*: High-level scorecard for drivers (Lando vs Oscar), showing total wins, podiums, DNFs, and average finishing position. This is ideal for executive dashboards.
2. *MV_Performance_By_Circuit_Type*: Directly addresses how drivers perform on different track types (*High-Speed, Technical, Balanced*). This helps strategists understand driver strengths and weaknesses and can inform car setup.
3. *MV_Wellbeing_Performance_Link*: Investigates the critical link between a driver's mental state and their on-track performance. It categorizes pre-race stress scores into levels (*Low, Medium, High*) and calculates the average finish position and DNF rate for each level.
4. *MV_Driver_Strategy_Impact*: A powerful view for strategic analysis. It aggregates performance outcomes based on combinations of weather, tire selection, and planned

pit stops, allowing the team to evaluate which strategies are most effective under specific conditions.

By creating these MVs, we provide Power BI with clean, purpose-built, and high-performance data sources, enabling the McLaren F1 management team to derive insights quickly and efficiently.

PowerBI

1. Driver Season Summary

Insights

- Highlights the total number of races, the total podiums as well as wins for each driver. This is beneficial as it allows managers to instantly compare the performance across teammates for an entirety of a season. This helps foster data backed discussions on consistency and areas for improvement.

Driver	Total Races	Total Podiums	Total Wins
Lando Norris	13	9	1
Oscar Piastri	13	3	0

Analysis

- The table shows that Lando Norris has had a hire podium finish rate at 9 in comparison to Oscar Piastri at 3 whilst also having one win under his belt. This can be used as a base for further analysis as to why Lando Norris has had a better season in totality in comparison to Oscar Piastri.

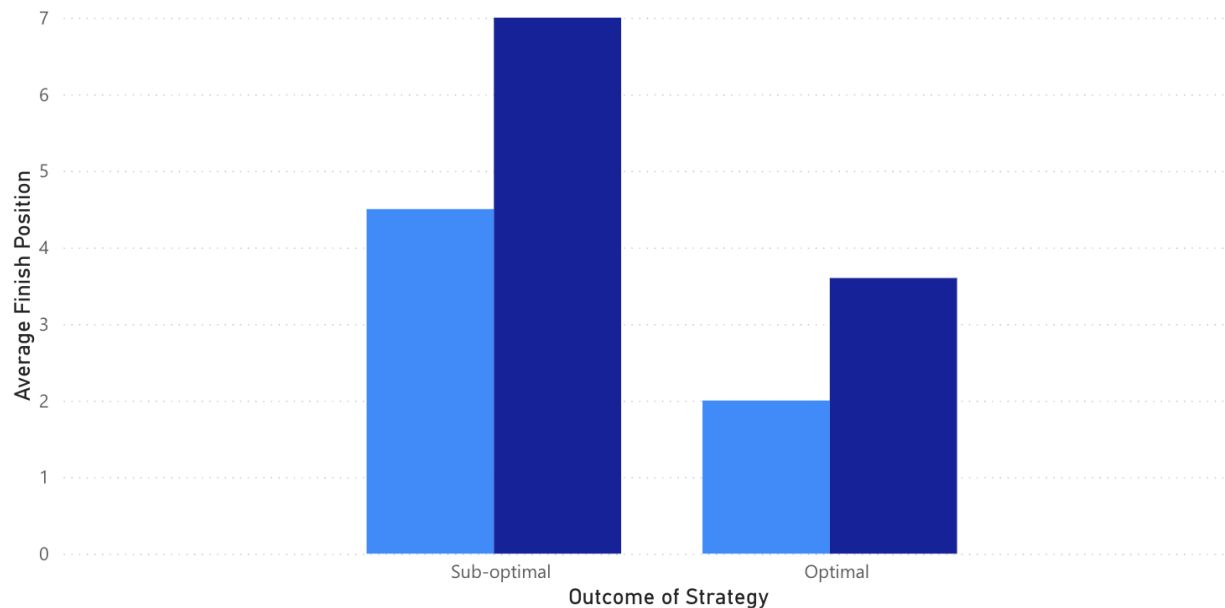
2. Driver Strategy Analysis

Insights

- Shows how the execution of different race strategies on a sub optimal and optimal level correlate to the average finish position. Not only that, it allows for quick identification of what weather conditions (e.g., hot weather vs wet tracks) bring out optimal strategy based on each driver.

Average Position Based on Outcome of Strategy

Driver ● Lando Norris ● Oscar Piastri



Analysis

- The bar chart demonstrates that an optimal outcome strategy for Norris almost always leads to a podium finish suggesting positive impacts when strategies are well executed. This contrasts Piastri's race strategies where despite optimal outcome of strategy, he still on average finishes outside the podium suggesting changes in strategies.

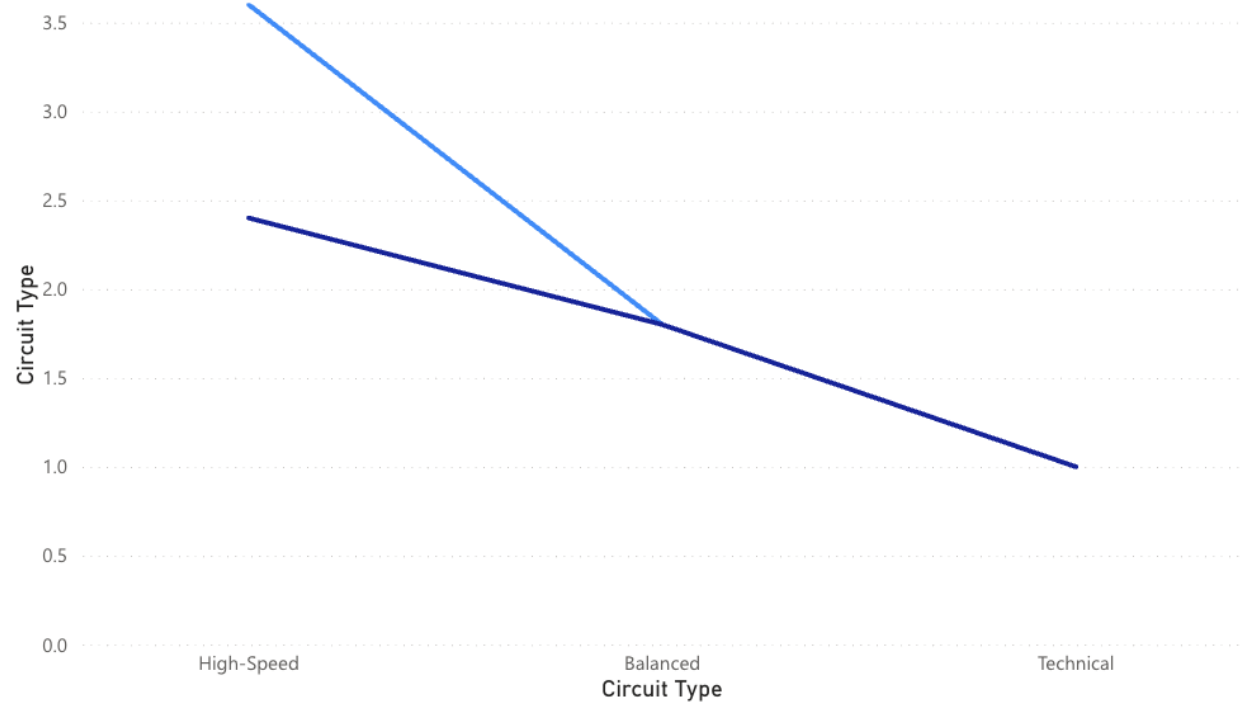
3. Performance by Circuit Analysis

Insights

- Highlights the driver performance trends on high speed, balanced and technical tracks. The data on the average finish position, overtakes, and tire degradation help engineers customise the car setups to suit the characteristics of the circuit.

Average Number of Overtakes by Circuit Type

Driver ● Lando Norris ● Oscar Piastri



Analysis

- Piastri performs better on high-speed tracks in comparison to Norris suggesting possible adjustments to Norris's car aerodynamics as well as training to increase performance. On average both drivers' worst performing tracks are high speed tracks. Thus, training tailored for high-speed tracks can be optimised for both drivers which could help increase performance on high-speed tracks.

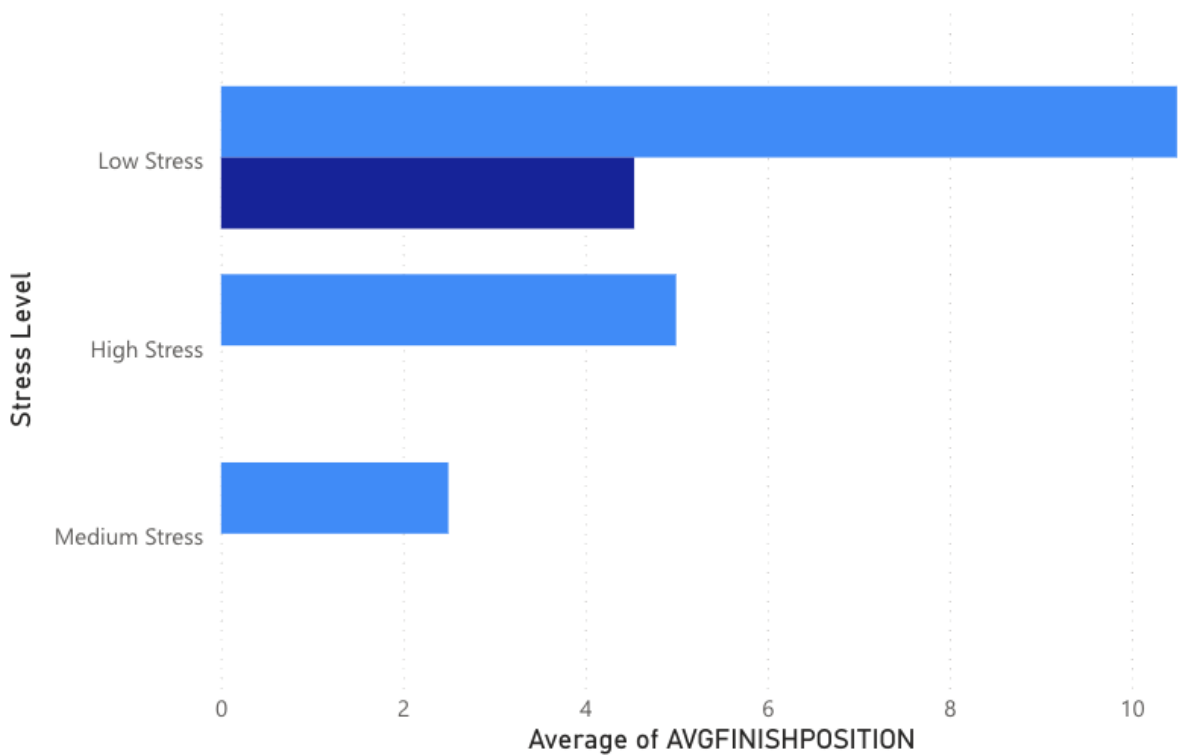
4. Driver Wellbeing Analysis

Insights

- Demonstrates how varying stress levels (low stress, high stress, medium stress) affects the performance of the driver. This is calculated through the number of DNF's based on the driver's stress levels as well as average finish positions. It would then allow management to correctly identify how important a certain stress level is for a driver to perform at his best.

Average Finish Position by Stress Level

Driver ● Lando Norris ● Oscar Piastri



Analysis

- Data shows Norris underperforms in low-stress scenarios likely due to a lack of competitive pressure. Norris performs at his best at medium stress levels, finishing on average on the podium. This data can be used to suggest ways to get Norris at medium stress levels to foster optimal performance.

Key Considerations

The development of this Power BI solution relies on several important considerations to ensure the usability, scalability as well as data accuracy:

Data Quality and Refresh Rates:

- The solution includes variables such as Stress Levels (low, medium, high) which are based on categorized quantitative data which in this case is heart rate to reduce ambiguity. The higher the heart rate the more likely it is medium – high stress levels. In addition, materialized views are constantly updated after each race in order to ensure accurate insights and reliable analytics.

Security and Data Access

- F1 data is highly sensitive, therefore access roles have to be defined to ensure that only authorized personnel can view certain driver health or performance data, ensuring confidentiality

Future Expandability:

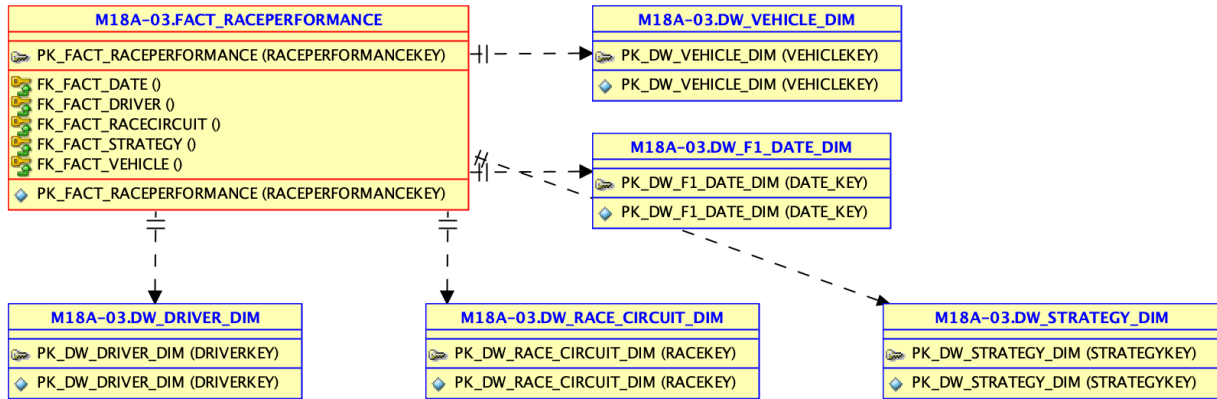
- The data model and Power BI setup is designed to easily scale and include data across multiple seasons and potentially allow for enhanced predictive analytics models in the future

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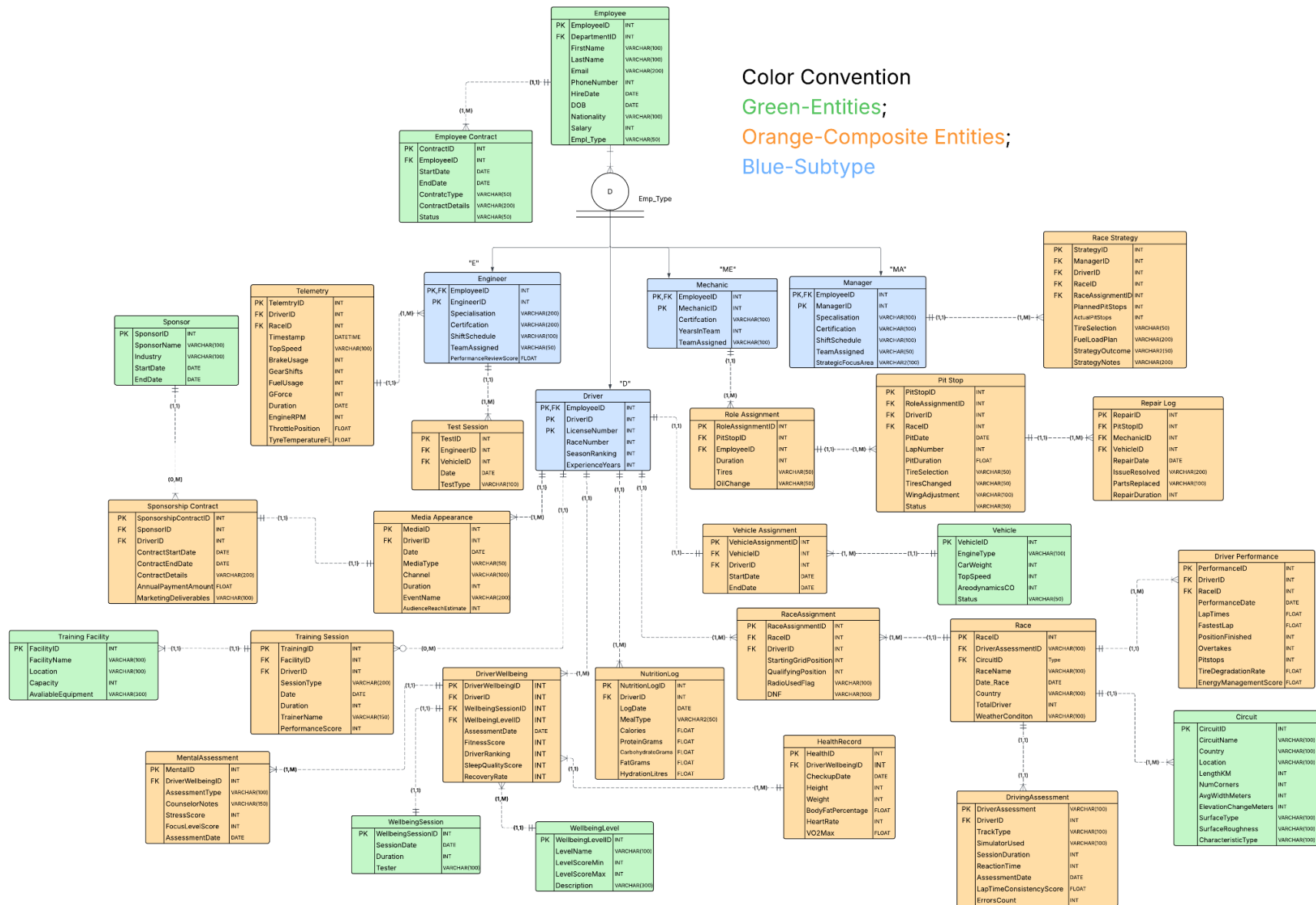
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Appendices

Data Warehouse Star Schema Diagram



Entity-Relationship Model



Star Schema

